

Labwork 4: Threads

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1 Introduction

This report presents an implementation of RGB image-to-grayscale conversion using both a traditional 2D CPU approach and a 2D CUDA-based GPU approach.

2 Method

An RGB image represents each pixel using three color components: red (R), green (G), and blue (B). The grayscale value is calculated using the weighted sum:

$$g(y, x) = \frac{1}{3} \sum_{i=0}^2 I(y, x, i) \quad (1)$$

3 CPU Implementation

The CPU implementation processes the image sequentially, converting each RGB pixel to grayscale while preserving the original 2D image structure rather than flattening it into a 1D array. For the test image, the 2D implementation takes 9.15 seconds, compared with 7.028 seconds for the 1D implementation, making the 2D approach approximately slower than the flattened 1D approach.



Figure 1: Original RGB image

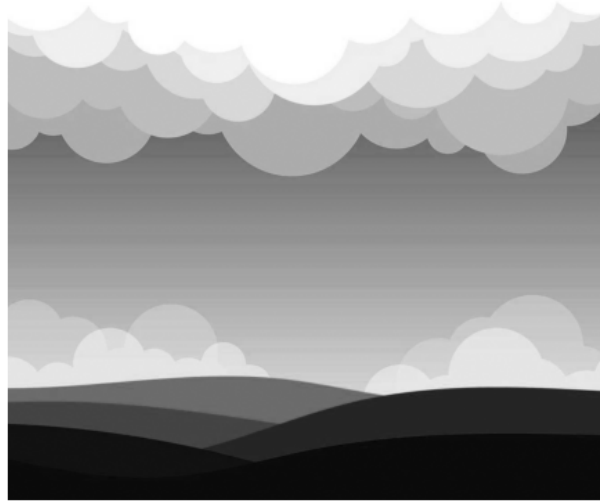


Figure 2: CPU Grayscale image

4 CUDA GPU Implementation

The second implementation uses CUDA through the Numba library to accelerate the grayscale conversion on the GPU. Unlike the previous lab, where the image was flattened into a 1D array, this implementation preserves the original 2D image structure. For the same test image, the CUDA implementation in 2D slightly slower than 1D, 0.129665 seconds compared to 0.094863 seconds.



Figure 3: GPU Grayscale image